

Initial Testing Run:

```
MyTrip_Final_Submission ./gradlew test --rerun-tasks
> Task :test
TripControllerTest > getExistingIdReturns200() FAILED
    java.lang.AssertionError at TripControllerTest.java:66
TripControllerTest > postValidTripReturns201() FAILED
    java.lang.AssertionError at TripControllerTest.java:37
TripControllerTest > putValidTripReturns200() FAILED
    java.lang.AssertionError at TripControllerTest.java:82
TripTest > testNegativeFuelPriceThrowsException() FAILED
    java.lang.IllegalArgumentException at TripTest.java:73
TripTest > testNegativeDistanceThrowsException() FAILED
    java.lang.IllegalArgumentException at TripTest.java:66
23 tests completed, 5 failed
> Task :test FAILED

[Incubating] Problems report is available at: file:///Users/angelicaalvarado/Desktop/sites/CS4398/MyTrip_Final_Submission/build/reports/problems/problems-report.html
FAILURE: Build failed with an exception.

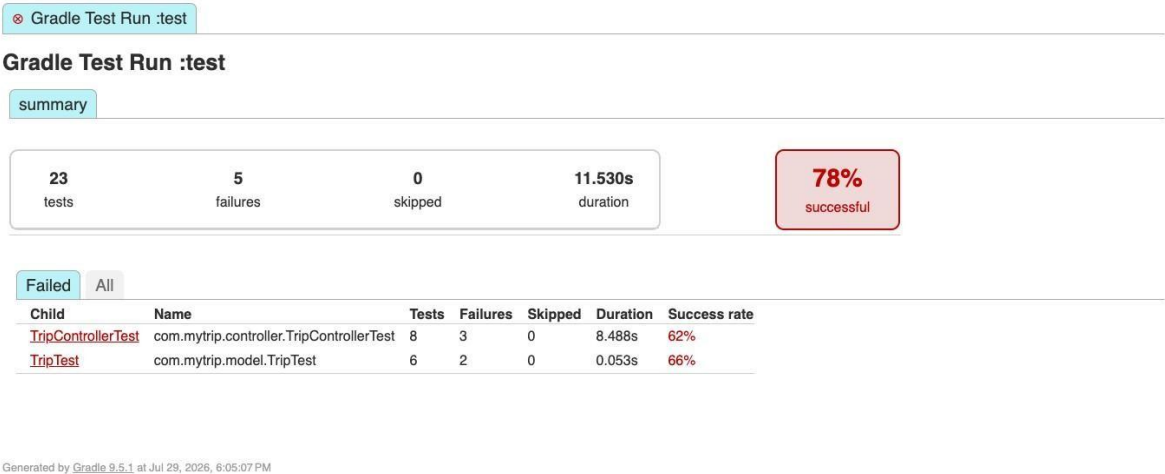
* What went wrong:
Execution failed for task ':test'.
> There were failing tests. See the report at: file:///Users/angelicaalvarado/Desktop/sites/CS4398/MyTrip_Final_Submission/build/reports/tests/test/index.html

* Try:
> Run with --scan to get full insights from a Build Scan (powered by Develocity).

Deprecated Gradle features were used in this build, making it incompatible with Gradle 10.
You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.
For more on this, please refer to https://docs.gradle.org/9.5.1/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD FAILED in 43s
4 actionable tasks: 4 executed
MyTrip_Final_Submission open build/reports/tests/test/index.html
```

All Results



All Results

⊗ Gradle Test Run :test

Gradle Test Run :test

summary

24

tests

6

failures

0

skipped

9.412s

duration

75%

successful

Failed

All

Child	Name	Tests	Failures	Skipped	Duration	Success rate
TripTest	com.mytrip.model.TripTest	6	3	0	0.037s	50%
MytripApplicationTests	com.mytrip.MytripApplicationTests	1	1	0	4.433s	0%
TripServiceTest	com.mytrip.service.TripServiceTest	9	2	0	0.288s	77%

Figure 2. Intermediate JUnit test run showing failures discovered during testing.

Trip Test: Expected java.lang.IllegalArgumentException to be thrown, but nothing was thrown.

The initial automated test run exposed problems in the model, service, and application-context tests. These failures were reviewed to determine whether they resulted from incorrect test assumptions or missing validation in the production code.

Third round of testing:

All Results

● Gradle Test Run :test

Gradle Test Run :test

summary

24

tests

0

failures

0

skipped

10.406s

duration

100%

successful

Child	Name	Tests	Failures	Skipped	Duration	Success rate
TripControllerTest	com.mytrip.controller.TripControllerTest	8	0	0	2.042s	100%
TripTest	com.mytrip.model.TripTest	6	0	0	0.058s	100%
MytripApplicationTests	com.mytrip.MytripApplicationTests	1	0	0	6.570s	100%
TripServiceTest	com.mytrip.service.TripServiceTest	9	0	0	0.411s	100%

all > com.mytrip.controller.TripControllerTest

● Gradle Test Run :test

TripControllerTest

summary standard output

8

tests

0

failures

0

skipped

2.042s

duration

100%

successful

Child	Tests	Failures	Skipped	Duration	Success rate
deleteExistingTripReturns204()	1	0	0	0.018s	100%
getAllTripsReturns200()	1	0	0	0.029s	100%
getExistingIdReturns200()	1	0	0	0.609s	100%
getMissingIdReturns404()	1	0	0	0.021s	100%
getOwnerSpecificTripsReturns200()	1	0	0	0.026s	100%
postInvalidTripReturns400()	1	0	0	0.246s	100%
postValidTripReturns201()	1	0	0	0.046s	100%
putValidTripReturns200()	1	0	0	0.017s	100%

all > com.mytrip.model.TripTest

● Gradle Test Run :test

TripTest

summary

6

tests

0

failures

0

skipped

0.058s

duration

100%

successful

Child	Tests	Failures	Skipped	Duration	Success rate
testExactlyOnBudgetTrip()	1	0	0	0s	100%
testNegativeDistanceThrowsException()	1	0	0	0.001s	100%
testNegativeFuelPriceThrowsException()	1	0	0	0.037s	100%
testOverBudgetTrip()	1	0	0	0.003s	100%
testStandardTripCalculations()	1	0	0	0.011s	100%
testZeroMpgThrowsException()	1	0	0	0.001s	100%

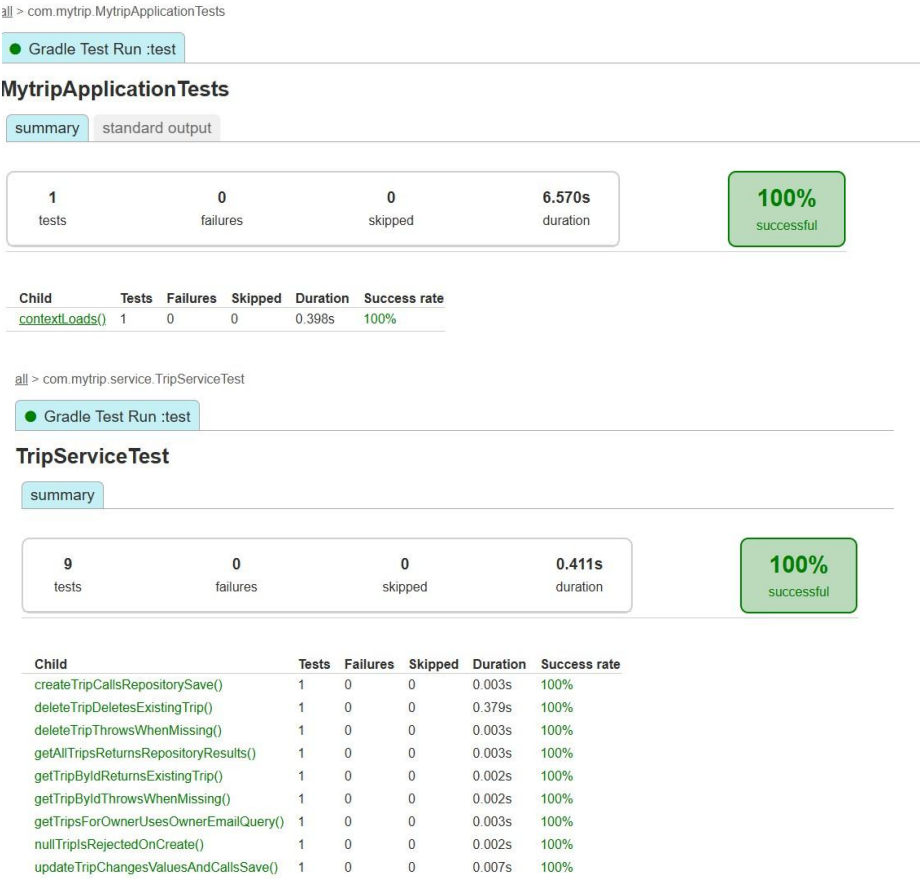


Figure 3. Final Gradle test report showing 29 tests, 0 failures, and 100% success.

After correcting the test setup and adding appropriate model validation, the complete Gradle test suite was executed again. All 29 automated tests passed with no failures or skipped tests.

TripControllerTest: 8 tests
TripServiceTest: 9 tests
TripTest: 11 tests
MytripApplicationTests: 1 test

Total: 29 tests
Passed: 29
Failures: 0
Errors: 0
Skipped: 0

IDE-Integrated AI Verification

During the final verification phase, GitHub Copilot Agent was used directly inside Visual Studio Code to inspect the source code, execute the Gradle test suite, diagnose verified defects, and confirm the final results.

Copilot-assisted verification identified and corrected validation behavior for zero-distance and zero-budget inputs and corrected Spring request-parameter binding through the Java `-parameters` compiler option.

The final automated suite contained 29 tests. All 29 passed with 0 failures, 0 errors, and 0 skipped tests. Exported Copilot chats and screenshots are included under the `ai-evidence` folder.